

1 Global Timbre Editor

Utterance-level timbre control

Reference Speech



Global Edit Instruction

“Make the voice warmer and breathier”

Global Timbre Editing



Trainable TTS Backbone

Edited Reference

Reusable Identity Anchor



Edit Once,
Reuse Across Segments

2 Local Multi-Instruction Generator

Segment-level expression control

Text & Instruction Timeline

Text (3 segments)

In a quiet village, the wind
moved through the trees,

and suddenly, a bright light
filled the sky!

Then everything
became still.

Local Instructions

calm and steady narration

excited and energetic emphasis

soft and gentle ending

KV-cache refresh at instruction switch

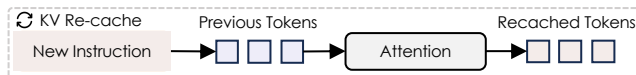
Rebuild conditioning while preserving acoustic continuity

shared edited
reference

Edited Reference
Shared across segments

Current Instruction
Active for current segment

Acoustic Context
Recent generated tokens



Generation over time

Segment 1
calm and steady narration

Segment 2
excited and energetic emphasis

Segment 3
soft and gentle ending

Frozen Autoregressive TTS Backbone

Generate with the refreshed KV-cache

EDICT: Edit the Voice Once, Then Direct It over Time

Timbre Anchor



calm and steady narration



excited and energetic emphasis



soft and gentle ending



- ✓ One continuous utterance
- ✓ Consistent edited timbre
- ✓ Time-varying expression